

SPARKFLOW

SparkFlow bridges the Sparkplug B stream Host decodes onto a NATS/JetStream backbone, so every point stays live and available company-wide, with no tag behind it, past the point where a tag-counted system stops scaling.

— VERSION 2.8.0

IGNITION 8.1.19+ AND 8.3

RUNS ON: THE HOST GATEWAY

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OVERVIEW

1. OVERVIEW

SparkFlow publishes every point Host decodes onto a NATS JetStream stream as a named, timestamped, quality-stamped message, independent of whether a tag exists for it. Dashboards, historians, analysts and AI pipelines subscribe directly; the plant is asked for nothing extra. A key-value store holds the current value and quality of every point, so a new subscriber gets the last known value of everything the moment it connects, and the stream keeps history so a window of past data can be replayed. SparkFlow also carries a command bridge: a NATS request/reply call can drive a confirmed NCMD/DCMD write back to the field.

SparkFlow is optional and is not part of the core SparkBridge install. It runs on the Host gateway, on its own perpetual per-gateway license, and is installed after Provider, Host and Edge are running.

REQUIREMENTS

IGNITION

JAVA

GATEWAY ROLE

INSTALL

3. INSTALL

1. Download the release bundle for your Ignition line from the [download page](#), including `SparkBridge-Flow.mod1` (optional add-on, bought separately from the core bundle).
2. Install after Provider, Host and Edge are running: Config > Modules > Install or Upgrade a Module, upload `SparkBridge-Flow.mod1`.
3. Accept the certificate and license once; confirm Config > Modules shows SparkFlow state = Running.

See `docs/FLOW_MODULE.md` in the bundle for the full configuration and lifecycle reference.

LICENSE

From release 2.7.0, SparkFlow runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). When the trial expires with no verified key naming the product, SparkFlow stops its dispatcher. A key dropped in after expiry lifts the pause on its own within the 30-second re-check, no button press and no restart. A verified key overrides the trial entirely.

CONFIGURE

5. CONFIGURE

All keys are `-D` system properties, read at startup as bootstrap/defaults. The `tagfeed` filter additionally hot-reloads from the `sparkflow-config` NATS KV bucket at runtime; nothing else currently hot-reloads.

SETTING	DEFAULT	MEANING
<code>sparkbridge.flow.enabled</code>	true	Master on switch. With false, start logs and run without connecting to NATS or to broker.

SETTING	DEFAULT	MEANING
<code>sparkbridge.flow.nats.url</code>	<code>nats://127.0.0.1:4222</code>	NATS core connect U
<code>sparkbridge.flow.tagfeed</code>	<code>empty</code>	Comma-separated NATS-sub style filter = one token remainder selecting subset pr onto the managed provider. E means no projects. I reloads fr sparkflc config K bucket.
<code>sparkbridge.flow.allowWildcardTagfeed</code>	<code>false</code>	Must be explicitly true before tagfeed fi list may co the whole estate wil uns.>.
<code>sparkbridge.flow.subjectprefix</code>	<code>uns.</code>	Telemetry subject pr Present fo future configurat the shippe

SETTING	DEFAULT	MEANING
		subject m currently hardcode: so changi key alone effect on wire form
sparkbridge.flow.kvbucket	uns-last	KV bucke holding ta last-know value and lifecycle s Created automatic absent.

SETTING	DEFAULT	MEANING
sparkbridge.flow.configbucket	sparkflow-config	KV bucket SparkFlow watches for own hot-reloadable config. Not created automatically must be provisioned ahead of time or absent at startup, SparkFlow throws an error and continues static bootstrap with hot-reload disabled.
sparkbridge.flow.twin.kvbucket	uns-model	Digital-twin bucket. Created automatically or absent, unless the config bucket.
sparkbridge.flow.commandTimeoutMs	5000	Timeout for reverse (N request/re NCMD/DC command
sparkbridge.flow.commandWorkers	8	Threads in command

SETTING	DEFAULT	MEANING
		bridge workers pool.
<code>sparkbridge.flow.commandQueue</code>	64	Bounded command queue in front of command workers. When full, a new command is not answered once with {"busy": true} rather than queued.
<code>sparkbridge.flow.commandPerEdge</code>	2	Commands allowed in parallel per group. A dead edge pins at most many workers.
<code>sparkbridge.flow.nats.credentialsFile</code>	none	NATS credential path, resolved through the secrets file preference bare -D value.
<code>sparkbridge.flow.nats.tls.enabled</code>	false	Whether to enable NATS connection TLS.

SETTING	DEFAULT	MEANING
sparkbridge.flow.nats.tls.certPath / keyPath / caPath	none	TLS cert/paths, res through tl secrets fil same way credential
sparkbridge.flow.cmd.keysDir	empty	Directory signed-command verificatio (public ke only, not r through tl secrets fil
sparkbridge.flow.cmd.requireSignature	false	When true unsigned command request is rejected if of falling l to the uns path.

Every subject, command subject and KV key includes a mandatory device slot: `uns.<group>.<edge>.<device|_>.<metric segment>[...]`, with a literal underscore meaning no device. Metric segment characters outside `[A-Za-z0-9_-]` are percent-style escaped as `=xx` (two hex digits); a literal `=` becomes `=3D`. This escaped form is canonical on the wire subject, the KV key, and the historian node key.

OPERATE

Confirm the Edge's Sparkplug traffic reaches NATS subjects under the `uns.` prefix, for example with a NATS CLI subscription to `uns.>`. Watch for repeated `{"busy":true}` command responses, which indicate the command queue or per-edge limit is saturated rather than a dead command bridge.

On a NATS link loss, SparkFlow's dispatcher stops publishing until reconnected; it does not buffer independently of what NATS/JetStream itself retains. On restart, SparkFlow reconnects to NATS and re-subscribes to Host's decoded stream; the tagfeed filter reloads from the config bucket if present.

A metric segment that is permanently unmappable for an unsupported wire value type or a rejected group/edge/device identifier publishes one event to `uns.$events.reject` and is never queued for retry.

VERIFY

7. VERIFY

Confirm the Edge's Sparkplug traffic reaches NATS subjects under the `uns.` prefix, for example with a NATS CLI subscription to `uns.>`. See `docs/FLOW_MODULE.md` in

the bundle for the subject naming contract and the KV bucket layout used for last-known values.

TROUBLESHOOT

SYMPTOM	LIKELY CAUSE	FIX
	<code>sparkbridge.flow.enabled</code> is false, or <code>tagfeed</code> is empty	Confirm the module is enabled and set a non-empty tagfeed filter
	<code>sparkflow-config</code> KV bucket was never provisioned	Provision the bucket ahead of time; SparkFlow will not create it itself
	Command queue full or per-edge limit reached on a dead/slow edge	Investigate the named edge; raise <code>commandWorkers/commandQueue</code> only after confirming the edge itself is healthy
	Pre-escaping defect in an old build, or an unsupported wire value type	Confirm the module escapes metric segments as <code>=XX</code> ; an unsupported value type instead routes to <code>uns.\$events.reject</code>

SYMPTOM	LIKELY CAUSE	FIX
	Module trial expired, no verified key	Drop a valid <code>sparkbridge-license.key</code> in the data directory

REFERENCE

9. REFERENCE

LICENSE FILE	<code><data>/sparkbridge-license.key</code>
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MODULE FILE	<code>SparkBridge-Flow.modl</code>
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DEFAULT NATS URL	<code>nats://127.0.0.1:4222</code>
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LAST-VALUE KV BUCKET	<code>uns-last</code>
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CONFIG KV BUCKET	<code>sparkflow-config</code> (not auto-created)
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DIGITAL-TWIN KV BUCKET	<code>uns-model</code>
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REJECT EVENT SUBJECT	<code>uns.\$events.reject</code>
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LICENSE RE-CHECK	30 seconds
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SUPPORT

Full install and config detail: [SparkBridge Docs, Flow](#). Product page: [SparkFlow](#).
Questions: [Contact](#). Release history: [GitHub releases](#).